3rd Milestone

Main idea

- Explain OrcaLink to people who may not know about whales or databases
- Focus on connecting family information and sighting records
- Make it clear that different records support different conclusions

Intro

- Start with the whales and the information we are studying
- Explain the goal before introducing technical details
- Define “matriline” as a maternal family group

Showing the existing sources

- Show what each source contributes
- CWR provides individual identity and family information
- Dryad links whale records to sightings
- The Whale Museum provides group sighting reports
- This comparison explains why combining the records needs care

Explaining the gap

- Some reports identify individual whales; others identify only groups
- Reporting a pod does not confirm every individual member
- Some whale identifiers come from group reports
- We need to preserve these differences when connecting records
- A shared sighting does not prove a close social relationship

Proposed model

- Use a simple diagram
- Show Whale and Matriline to explain family membership
- Show Kinship Assertion separately from Association
- Separate Individual Observation from Group Observation
- Connect observations to Sightings
- Include Data Source so the evidence behind records remains traceable
- Leave detailed attributes and participation limits out of the presentation sketch

Why it matters

- Researchers could examine family relationships alongside repeated sightings
- They could check which records support a connection
- Keeping identification information helps them interpret results appropriately
- Describe these as intended benefits because the model has not yet been evaluated

Knowledge graph

- Describe it as connected records with labeled relationships
- Explain the questions we aim to answer:
    - Family connections
    - Repeated co-sightings
    - Indirect connections

Outro

- Remind the audience that this is a proposed model
- Describe potential associations as candidates for further investigation
- End with the goal
    - Making records easier to connect while keeping their meaning and evidence clear

Our project, OrcaLink, asks how we can connect records about Southern Resident killer whales to explore their potential social ties. These whales have family relationships, but records of their families and records of sightings describe different things.

Our goal is to bring those records together while keeping the evidence behind each connection clear. This could help researchers examine family relationships alongside repeated sightings, and trace a connection back to the records supporting it.

Whale recording has developed from handwritten logs and film photographs to digital identification and databases. Our review compares sources by what they represent. The Center for Whale Research describes individual identities and family groups. The Dryad dataset links whale records to sightings. The Whale Museum records reported groups, dates, and locations.

Each source is useful, but their level of detail differs. A group report might name a pod without identifying every whale. Some individual identifiers in the Dryad data come from group reports. So an identifier alone does not always mean someone individually confirmed that whale.

There is another distinction: two whales recorded together are not necessarily close social partners. Our modeling challenge is to connect family and sighting records without losing these differences or overstating what the evidence shows.

In the diagram, whales connect to their maternal family groups, called matrilines. Kinship assertions record relationships between whales. Sightings connect to individual observations and group observations separately. This lets us keep a report even when the individual whales are unknown.

Associations represent recorded co-sightings or potential connections. Their type, method, and time window help distinguish the results. Source information preserves where the observations and relationship evidence came from.

We plan to develop this conceptual model into a relational database, then a knowledge graph. The graph will represent whales and records as connected nodes with labeled relationships. We aim to query family connections, repeated co-sightings, and indirect connections while checking their supporting evidence.

OrcaLink is still a proposed model. A potential connection would be a candidate for further investigation, not proof of friendship. The aim is to make the records easier to connect and the meaning of each connection easier to examine.

Sources:

- Team milestone 2 report
- 02_Background_Aung_Bowen.pdf
- CWR Orca Identification: https://www.whaleresearch.com/orca-id
- Dryad doi:10.5061/dryad.gf1vhhn2f

- The Whale Museum supplied dataset